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D212 – Task 1

Performance Assessment

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PART 1: Research Question

A. The Purpose of the Data Mining Report

1.

Can the medical and demographical information of the patients be used to group the vitamin D levels using the k-means clustering analysis?

2.

The analysis aims to see if the variables in the data set can be used to group the patients based on their vitamin D levels. Since k-means clustering only uses continuous variables, the data set will be limited to exclude all categorical data.

Suppose the data can be used to group the patients based on their vitamin D levels. In that case, further research can be done to determine if certain health conditions or demographical information is more prevalent in one group.

This will allow doctors to understand their patients better and determine their treatment plans' best course of action.

PART II: Method Justification

B. Classification Method

1.

K-means clustering is a popular unsupervised machine learning algorithm used to partition a dataset into distinct groups (clusters) based on feature similarity. Here's how it analyses the selected dataset (Anonymous, 2024):

- a) Initialization: The algorithm randomly selects (k) initial cluster centroids.
- b) Assignment: Each data point is assigned to the nearest cluster centroid based on the Euclidean distance.
- c) Update: The centroids of the clusters are recalculated as the mean of all data points assigned to each cluster.
- d) Iteration: Steps 2 and 3 are repeated until the centroids no longer change significantly, indicating stable clusters.

Expected Outcomes for a k-means clustering analysis on the vitamin D levels of the patients:

- Cluster Formation: The algorithm will partition the individuals into three clusters based on their vitamin D levels. Each cluster will represent a group of individuals with similar vitamin D levels.
- Cluster Characteristics: Each cluster will have a distinct range of vitamin D levels. For example:
 - Cluster 1: Low vitamin D levels
 - Cluster 2: Medium vitamin D levels
 - Cluster 3: High vitamin D levels

2.

K-means clustering makes several key assumptions (Everitt, Landau, Leese, & Stahl, 1984):

1. **Spherical Clusters:** It assumes that clusters are spherical and equally sized, meaning that data points in each cluster are distributed evenly around the centroid.
2. **Linear Separability:** K-means assumes that clusters can be separated linearly, which may not hold true for all datasets.
3. **Homogeneity of Variance:** It assumes that all clusters have similar variances, which can impact the effectiveness of clustering.
4. **Number of Clusters:** The number of clusters (k) must be specified in advance, which can be a limitation if the optimal number is unknown.
5. **Continuous Features:** K-means works best with continuous numerical data, as it relies on calculating distances between data points.
6. **Independence of Clusters:** It assumes that clusters are independent and not overlapping.
7. **Initialization Sensitivity:** The algorithm assumes that the initial placement of centroids can influence the final clustering outcome, which may lead to different results with different initialisations.

3.

This assignment was completed using the Python environment. Python has a variety of packages and libraries that were specifically created for data science processes like modelling and predictions. The packages and libraries in Python work together seamlessly and the syntax is relatively straightforward and concise, making it easy to understand and implement. The internet is also full of resources surrounding the different packages and include information about how to use them, how to solve problems or errors you might encounter, different examples of how to get to the same results, and many more.

Some of the libraries and packages used for this project include:

- Pandas: Data manipulation and analysis library, providing data structures and functions for working with structured data.
- NumPy: Numerical computing library, supporting multi-dimensional arrays and matrices, along with a collection of mathematical functions.
- Seaborn: Statistical data visualization library, built on Matplotlib, for creating attractive and informative statistical graphics.
- Matplotlib: Plotting library, offering a wide range of static, animated, and interactive visualizations in Python.
- Sci-kit learn: Machine learning library, featuring various classification, regression, clustering algorithms, and tools for model fitting, data preprocessing, model selection, and evaluation.
- Seaborn: Statistical data visualization library, built on Matplotlib, for creating attractive and informative statistical graphics.

```
import seaborn as sns
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn import preprocessing
```

```
from sklearn.cluster import KMeans
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.metrics import silhouette_score
```

PART III: Data Preparation

C. The Data Preparation Process

1.

It is crucial to normalize the data in k-means clustering to ensure that all features contribute equally to the distance calculations in k-means clustering, leading to more meaningful and accurate clusters (Sharma, 2019). Since k-means clustering best suits continuous variables, the data set was limited to excluding all categorical variables. The table below shows the information for the data set intended for the k-means clustering analysis.

Variable	Description	Type	Format	Examples
Lat	GPS coordinates of patient residence as listed on the billing statement	Continuous	Float	34.3496
Lng	GPS coordinates of patient residence as listed on the billing statement	Continuous	Float	-86.72508
Income	Annual income	Continuous	Float	55586.48
Vit D Levels	ng/mL measure vitamin D levels	Continuous	Float	16.49455
Initial Days	Number of days in hospital for first visit	Continuous	Float	5.95725, 1.012586
Total Charge	Daily costs of patient's stay	Continuous	Float	2990.086, 3316.4
Additional Charges	Other charges during stay	Continuous	Float	18518.04, 10575.43

2.

Step 1: import all packages and libraries used in this analysis (see Part II Section B.3)

Step 2: import the data set and convert to a data frame.

```
data = pd.read_csv('C:\\Users\\LorraineFerrusi\\OneDrive - The Information Lab\\Documents\\Personal\\WGU\\D212\\Data\\medical_clean.csv')
```

Step 3: inspect the data frame - check for null values and duplicates.

```
print(data.head())
print(data.info())
print(data.isnull().values.any())
duplicates = len(data) - len(data.drop_duplicates())
print('There are ' + str(duplicates) + ' duplicates in this data set')
```

Step 4: only include continuous variables

```
data_clean = data.select_dtypes(include = ['float64'])
```

Step 5: normalize the data

```
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data_clean)
print(data_scaled)
```

Step 6: convert numpy array to dataframe to export as csv

```
data_transformed = pd.DataFrame(data_scaled, columns = data_clean.columns)
```

Step 7: export to csv

```
data_transformed.to_csv('C:\\Users\\LorraineFerrusi\\OneDrive - The Information Lab\\Documents\\Personal\\WGU\\D212\\Data\\df_transformed.csv')
```

3.

See attached CSV file (df_transformed.csv).

See attached Python file for data cleaning code (D212 Task 1 Data Cleaning.py).

PART IV: Analysis

D. Perform Data Analysis and Report on the Results

1.

Determining the optimal number of clusters in k-means clustering is crucial for obtaining meaningful results. In this analysis, the elbow method was used. The Elbow Method is a technique used to determine the optimal number of clusters in k-means clustering by plotting the within-cluster sum of squares (WCSS) against a range of cluster counts KKK. As KKK increases, WCSS decreases, but at a certain point, the rate of decrease slows, forming an "elbow" in the plot. This elbow point indicates the optimal KKK, suggesting that adding more clusters beyond this number yields diminishing returns in variance reduction. While effective, the method can be somewhat subjective, requiring visual inspection to identify the elbow point.

```
# Determine the optimal number of clusters using the Elbow Method
```

```
wcss = []
```

```
for i in range(1, 11):
```

```
    kmeans = KMeans(n_clusters = i, init = 'k-means++', max_iter = 300, n_init = 10, random_state = 0)
```

```
    kmeans.fit(vitd_scaled)
```

```
    wcss.append(kmeans.inertia_)
```

```
# Plot the Elbow Method graph
```

```
plt.plot(range(1, 11), wcss)
```

```
plt.title('Elbow Method for Vitamin D Levels')
```

```
plt.xlabel('Number of clusters')
```

```
plt.ylabel('WCSS')
```

```
plt.show()
```

2.

See attached Python file (D212 Task 1 k-means Clustering.py)

PART V: Data Summary and Implications

E. Summarize Findings

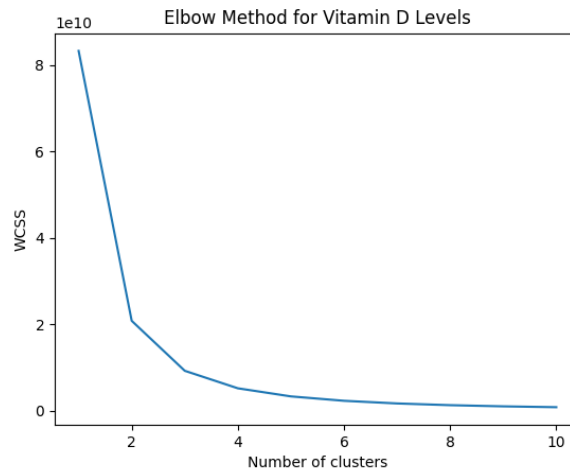
1.

To evaluate the quality of the clusters created by k-means clustering, there are several factors to consider:

- Within-Cluster Sum of Squares (WCSS):

WCSS measures the variance within each cluster. Lower WCSS values indicate that the data points within each cluster are closer to the cluster centroid, suggesting more compact and well-defined clusters.

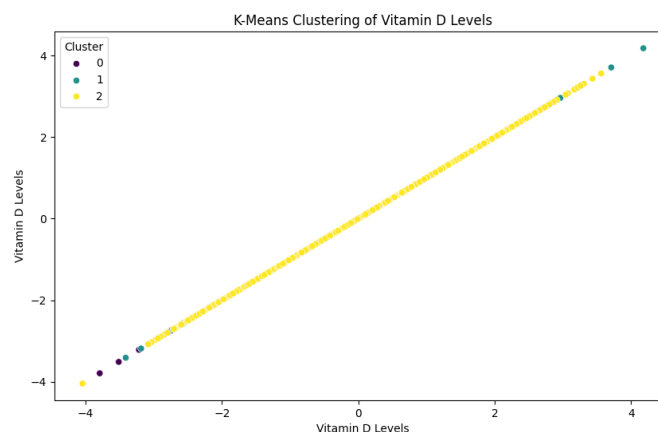
In the Elbow Method, there is a "knee" point where the WCSS decreases slower, indicating an optimal number of clusters. For this analysis, three clusters were chosen based on this method.



- Cluster Separation:

Good clusters should be well-separated from each other. This means that the distance between the centroids of different clusters should be larger than the distance within each cluster.

The scatter plot clearly shows that the clusters are not well-separated or distinctly grouped.



- Silhouette Score:

The silhouette score measures how similar a data point is to its own cluster compared to other clusters. It ranges from -1 to 1, with higher values indicating better-defined clusters.

A silhouette score close to 1 means that the data points are well-clustered, while a score close to 0 indicates overlapping clusters, and a negative score suggests that data points may have been assigned to the wrong cluster.

For this analysis, the silhouette score was 0.590, which indicates a moderately good clustering result.

- Cluster Size:

The size of each cluster should be balanced. If one cluster is significantly larger or smaller than the others, it may indicate that the clustering algorithm is not capturing the underlying structure of the data well.

For this analysis, the cluster sizes were 3340, 3334, and 3326, showing a balanced distribution among the clusters.

2.

The clustering analysis on vitamin D levels resulted in 3 clusters of similar size. The silhouette score of 0.590 indicates moderately well-defined clusters. However, the clusters are not clearly defined or separated. This could mean that there is overlapping in the clusters because the features used for clustering may not have clear boundaries, or the selected features might not be the best for distinguishing between clusters.

The centroids of the three identified clusters are as follows:

- Cluster 1: Characterized by slightly below-average vitamin D levels, significantly higher income, and lower initial days, total charges, and additional charges.
- Cluster 2: Characterized by near-average vitamin D levels, slightly below-average income, higher initial days, total charges, and near-average additional charges.
- Cluster 3: Characterized by slightly above-average vitamin D levels, lower income, significantly lower initial days, total charges, and slightly above-average additional charges.

The implications of these results are listed below:

- Cluster 1 may represent wealthier patients with fewer hospital stays and lower overall charges.
- Cluster 2 may represent patients with moderate income and higher hospital utilisation.
- Cluster 3 may represent lower-income patients with minimal hospital stays and charges.

These insights can help doctors tailor interventions and resource allocation based on patient group characteristics. However, additional steps are required to ensure that the analysis is accurate. At this stage, the health and demographical information in the data set is insufficient to clearly identify groups within the vitamin D levels.

3.

One limitation of this data analysis is the potential for feature selection bias. The clustering was performed using a specific set of features. If these features do not capture the underlying structure of the data well, the resulting clusters may not be meaningful. This can lead to misleading interpretations and conclusions

4.

Given the overlapping clusters and the moderate silhouette score, here are some recommendations for doctors using this data to group patients based on their vitamin D levels:

- Feature Engineering: Identify and include relevant features that may influence vitamin D levels, such as dietary habits, sun exposure, and medical history.
- Data Preprocessing: Ensure data is clean, normalised, and free of outliers to improve clustering accuracy.
- Exploratory Data Analysis (EDA): Perform EDA to understand the distribution and relationships between variables.
- Advanced Clustering Techniques: Consider using advanced clustering techniques like hierarchical clustering or Gaussian Mixture Models (GMM) for potentially better results.
- Validation: Validate the clusters using domain knowledge and additional metrics like the Davies-Bouldin Index or Dunn Index.
- Iterative Approach: Continuously refine the model by incorporating feedback from medical professionals and updating the features and clustering methods as needed.

References

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- Sharma, P. (2019, August). *Why is scaling required in KNN and K-Means?* Retrieved from Medium:
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- Winn, J., Bishop, C. M., Diethe, T., Guiver, J., & Zaykov, Y. (2020). K-Means Clustering. In J. Winn, *Model Based Machine Learning* (p. Chapter 8).